Homework - 1

CSL-100/MAL-400 Introduction To Programming

Question 1

Design a program to calculate the volume of a cuboid, cylinder, and sphere. Input:

- In line 1, l w h parameters for cuboid
- In line 2, r h parameters for cylinder
- In line 3, the r parameter for the sphere Output:

Print volumes of each on a separate line. (1st output for cuboid, 2nd line for cylinder, 3rd for sphere)

Input Format
Length, breadth and height of the cuboid Radius and height of cylinder Radius of sphere

Constraints
Restrict the output to 2 decimal places and take pi = 3.14

Output Format
Volume for cuboid
Volume for cylinder
Volume for sphere

Sample Input 0
3 2 5
4 6
4

Sample Output 0
30.000000
301.440002
267.946686

Question 2

Write a program to take n digit number as input from the user and reverse the number. Print the reversed number as output.
In the output, the line should contain the reversed number.
Note: Number Should not end with zero and if it happens the program should display "Error: Ending with zero."

Input Format
Take a positive integer.

Constraints
The number should not end with zero

Sample Input 0
Enter the Number:
51234

Sample Output 0
43215

Sample Input 1
56780

Sample Output 1
Error: Ending with zero.

Question 3

Given marks of the course for a student, follow the below grading scheme and print the corresponding grade as output. Constraints.
Note: Use only switch case.

Input Format
A float number represents the marks of the student.

Constraints
The number should not be negative and $0 \leq$ marks ≤ 100.

86 - 100: A+
71 - 85: A-
56 - 70: B+
41 - 55: B-

26 - 40: C
0 - 25: F

Sample Input 0
Enter the Number:
84.25

Sample Output 0
A-

Question 4

Consider the quadratic equation ax^2 + bx + c, you are given coefficients a, b and c as inputs, find and print the roots of the quadratic equation as output. If given coefficients does not form a quadratic equation, print "Not a Quadratic Equation" as output.

Note: D = b^2 - 4ac
If D $<$ 0 output both roots separated by space
If D = 0 output single root
If D $<$ 0 print "No Real Roots" as output

Input Format
3 values for a, b, c in a single line separated by spaces.

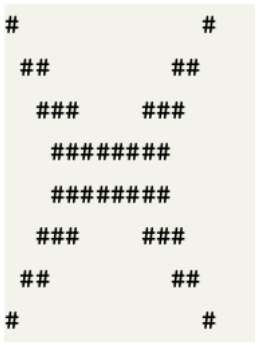
Constraints Restrict output to 3 decimal places

Sample Input 0
Enter the value of a, b and c:
2 5 -3

Sample Output 0
-3.000 0.500

Question 5

Write a C program that takes a positive integer n and prints a pattern like this: (given example forn = 4)



Question 6

Write a C program that takes a positive integer n and prints the first n^2 prime numbers in a square. (given example for n = 5).

Note : that its perfectly fine if the output is not exactly a square. It just

```
2 3 5 7 11
13 17 19 23 29
31 37 41 43 47
53 59 61 67 71
73 79 83 89 97
```

needs to have n primes in each row.

Question 7

Given two positive integers a and b ($a \leq b \leq 1000$), find the number of pairs (i, j) ($a \leq i \leq j \leq b$) such that i divides j.

Input Format
Enter the value of a and b.

Constraints
Both a and b are positive, $a \leq b \leq 1000$, and $a \leq i \leq j \leq b$

Sample Input 0
1
10

Sample Output 0

27

Sample Input 1
1
1000

Sample Output 1
7069

Question 8

Given a positive integer n ($n \leq 100$), find the number of positive integers less than or equal to n, that are expressible as sum of squares of two (not necessarily distinct) non-negative integers a and b.

Input Format
Enter a positive integer n.

Constraints
$n \leq 100$ and squares of two non-negative integers is not necessarily distinct.

Sample Input 0
5

Sample Output 0
4
as 1= $0^2 + 1^2$, 2=$1^2 + 1^2$, 4= $0^2 + 2^2$, 5= $1^2 + 2^2$

Sample Input 1
50

Sample Output 1
24

Question 9

Write a C program to find the sum of all numbers divisible by 17 within 100 to 999. It is easy with the modulus(%) operator. Can you do it without using the modulus operator?

Question 10

Check if the input integer can be expressed as the sum of two distinct prime numbers.

Input Format
Enter a positive integer n.

Constraints
Input is a positive integer and the output is true or false as per the case.

Sample Input 0
5

Sample Output 0
true// 5 = 3 + 2

Sample Input 1
7

Sample Output 1
true // 7 = 5 + 2

Submission

Please submit your homework in piazza under hw1 folder and make it a private submission to the instructors (not individual TA's or Subidh Sir). Zip all the codes and name the zip as **yourname_rollno**.

The submission deadline is 8:00 pm September 10.